

GEOG 5203

Geographic Information Science: Spatial Modeling

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	4
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (79.5 / 100)	62.6 / 100

63

CURRENT AVG SCORE
across 6 job roles

94

POTENTIAL SCORE
with recommended additions

+43

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Focuses on the use and development of advanced models for human and environmental applications in a geospatial environment integrating raster and vector data models. Covers terrain and hydrologic modeling, geostatistical modeling, dasymetric modeling, as well as multi-criteria modeling. Group projects critically design, implement and test spatial models to develop independent skillsets in a chosen problem setting.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	79.5	100.0	+20.5	MEDIUM
AI Engineer	65.6	100.0	+34.4	HIGH
ML Engineer	63.1	83.2	+20.1	MEDIUM
MLOps Engineer	59.8	89.3	+29.5	MEDIUM
Applied Scientist	56.7	100.0	+43.3	HIGH
Data Scientist	51.0	91.0	+39.9	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

4 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

79.5 → 100.0 (+20.5 pts)

sets learning analytics tools solutions exploratory business science

AI Engineer

65.6 → 100.0 (+34.4 pts)

perception learning build software technical model systems platform

ML Engineer

63.1 → 83.2 (+20.1 pts)

learning machine systems model world build problems technical

MLOps Engineer

59.8 → 89.3 (+29.5 pts)

model systems learning machine training tools

Applied Scientist

56.7 → 100.0 (+43.3 pts)

biology computational research storage interdisciplinary problems technical learning

Data Scientist

51.0 → 91.0 (+39.9 pts)

sets analytics presentation social build problems approaches

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.